

2] Set Comprehension - Set Comprehension

Creates a new set by iterating over a sequence & applying an expression to each element like list

Comprehension but we use curly braces $\{ \}$ & no duplicates in the result.

Syntax :-

$\{ \text{expression for element/item in sequence/iterable if condition} \}$

eg. Create a set having squares of numbers from 1 to 5.

exp. of - $\{ 1, 4, 9, 16, 25 \}$

Square-set = $\{ i ** 2 \text{ for } i \text{ in } \dots \}$

range(1, 6, 1) &

print(Square_set1)

* * - exponentiation

$$1 \text{ ** } 2 = 1^2 = 1 \quad 3 \text{ ** } 2 = 9 \quad 5 \text{ ** } 2 = 25$$

$$2 \text{ ** } 2 = 2^2 = 4 \quad 4 \text{ ** } 2 = 16$$

→ {1, 4, 9, 16, 25}

3] Dictionary Comprehension -

dictionary comprehension creates a new

dictionary by iterating over a sequence,

— producing key-value pairs using

key-expression & value expression,

with an optional condition to

filter entries

Syntax:

{ key expression : value expression

for element/item in sequence/iterable
if condition }

eg. create a dictionary having
Squares of no. from 1 to 5,
where key is no. & value is
square.

exp out - { 1: 1, 2: 4, 3: 9,
4: 16, 5: 25 }

Sq-dict = { i : i * * 2
for i in range(1, 6, 1) }

print (sq-dict)

O/P - { 1: 1, 2: 4, 3: 9,
4: 16, 5: 25 }

Q. 2 create a dictionary where
key is name & value

key is the animal name & value
is the length of animal name

animals = ["cat", "dog", "fish"]
 3 3 4

exp.out $\Rightarrow$

{ "cat": 3, "dog": 3,
 "fish": 4 }

animals = ["cat", "dog", "fish"]

dict_animals = { i : len(i)

for i in animals }

print(dict_animals)

o/p - { "cat": 3, "dog": 3, "fish": 4 }